

APPLICABILITY OF ADVANCED TECHNOLOGY IN SUPERORDINATE INSPECTION AND CONDITION MONITORING FOR STEEL PLANT EQUIPMENTS

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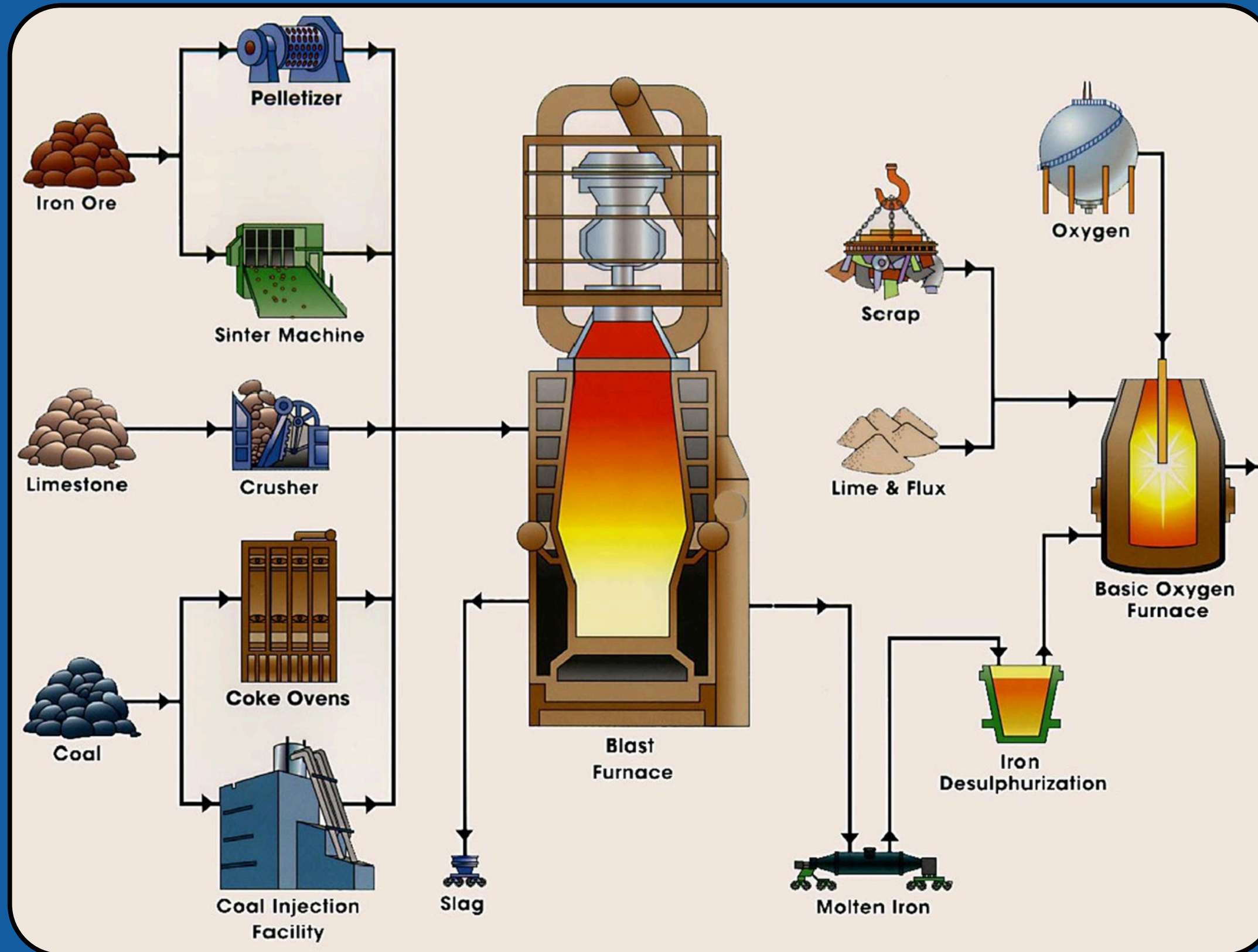
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PROJECT OBJECTIVE



STEEL MAKING PROCESS

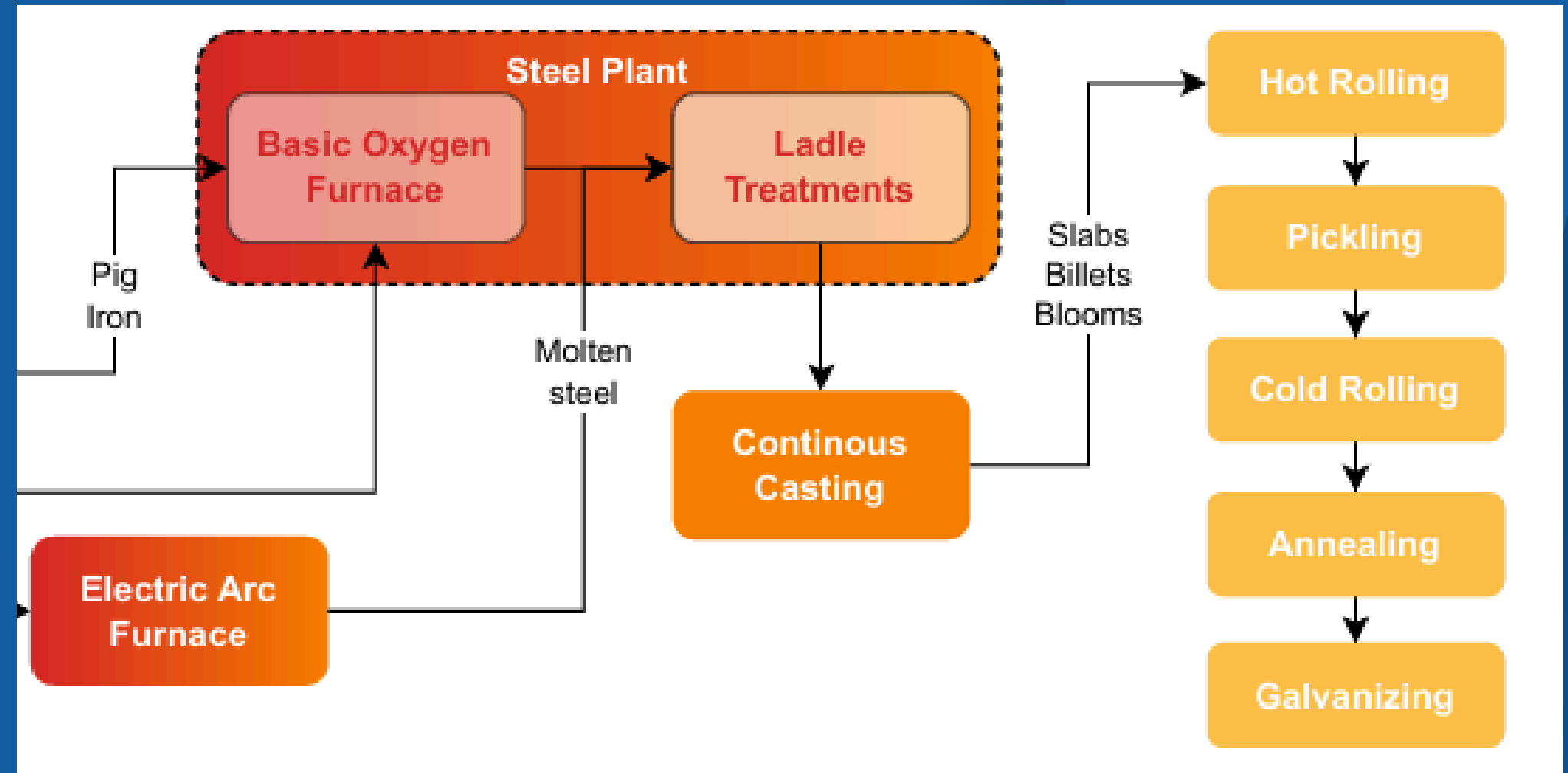
- **Raw materials used:** Iron ore, coke (from coal), and limestone.
- **Preparation of iron ore:** Pulverised with sand and impurity removed by magnetic separation.
- **Blast furnace:** The furnace is coated with refractory bricks to withstand high temperatures.
- **Loading the furnace:** Iron ore, coke, and limestone are added from the top of the furnace.

Formation of products:

The molten iron collects as pig iron (or cast iron) and the lighter slag (calcium silicate) floats above is removed.

Basic Oxygen Process

- Oxygen is blown over molten pig iron in blast furnace, oxidizing carbon and other impurities.
- Carbon content is greatly reduced, producing steel with much lower carbon, and slag containing impurities is removed.



Electric Arc Furnace (EAF)

- Uses electrical arcs between graphite electrodes and the metal charge in furnace.

“Another flaw in the human character is that everybody wants to build and nobody wants to do maintenance.” — Kurt Vonnegut

NEED FOR MONITORING

Potential Equipment Failures and Causes

Furnace & Process Safety

- Blast Furnace Hood / Cooling Leaks & Blocks: Water leaks risk hydrogen explosions and refractory burn-through.
- A BF water leak once caused 6 deaths and ¥27M losses, proving the need for leak/blockage monitoring .

Converters : Vibration stresses drivetrain and affects steel quality.

- In a BOF study over ~300 blows, vibration + audio features correlated strongly with process variables, enabling predictive control cues

Mills & Continuous Casting

- Mill Gearboxes: Gearbox breakdown halts rolling for days, scrapping tons of steel.
- A shaft failure caused 3 days outage and 1,800 t loss before gearbox replacement .
- Failure analysis of a PCI mill found grinding-roller bearing problems causing unplanned outages that increase production cost.

Continuous Casting Bearings: Roll-segment bearing failures force slowdowns and quality.

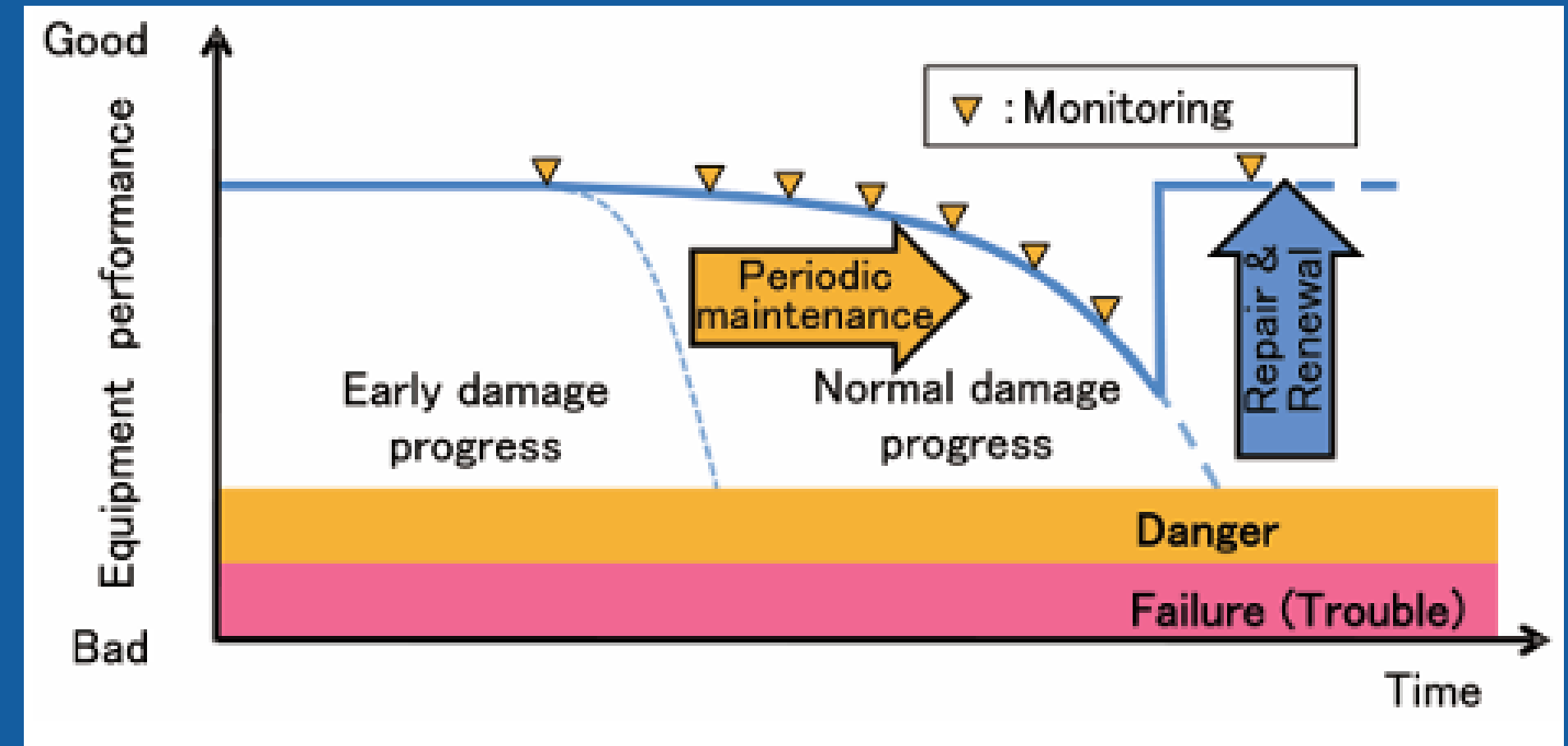
- Caster bearings often last just 1–2 months without proper vibration/lubrication monitoring .

Rotating & Utility Equipment

- Fans, Pumps, Bearings: Failures in ID fans, exhaust fans, and pumps cause sudden downtime.
- Around 80% of bearing failures come from poor lubrication (wrong type, contamination, starvation).

Material Handling Systems

- Conveyors (Pulley/Belt Failure): Stops material flow, causes spillage and damages gearboxes.



1. According to Siemens' True Cost of Downtime 2022 report, unplanned downtime now costs Fortune Global 500 companies about **\$1.5 trillion per year, which is ~11% of their revenues.**
2. The same report shows an average large manufacturing plant loses **\$129 million/year** due to unplanned downtime.

Predictive Maintenance: A Growing Priority

Steel plants face extreme conditions where failures are costly, the sharp rise in predictive maintenance adoption shows its growing need for safety, efficiency, and reliability.

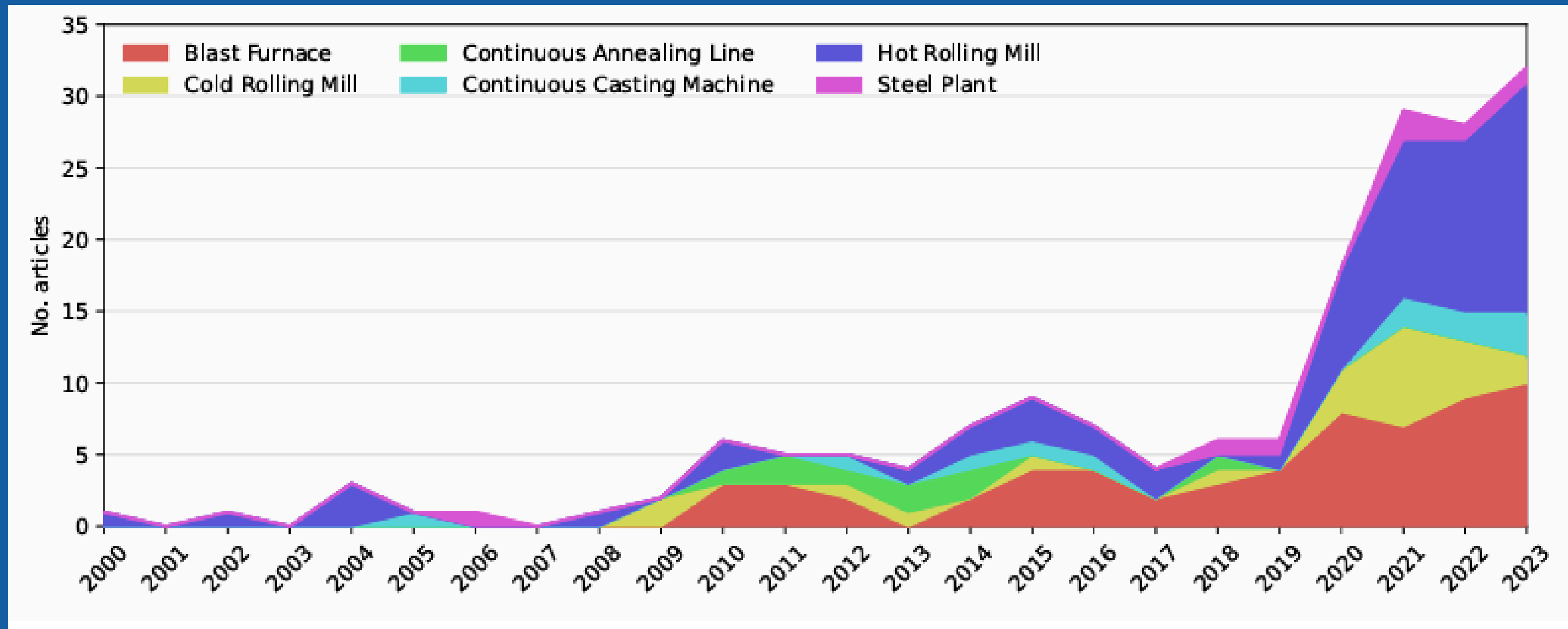


Figure: Historical distribution of installations subjected to predictive maintenance

- Implementation of PdM across manufacturing shows reduction in **unplanned downtime by up to 50%, and maintenance cost reduction by 18–25%.**

DIFFERENT PDM TECHNOLOGIES

PdM Technology	Global Best Practice	The Indian Reality (Gap)
Vibration Analysis & RUL	Highly automated Condition-Based Monitoring (CBM) systems are standard.	Dependence on Preventive Maintenance (time based) or Run to Failure (until breakdown)
Advanced Data Science (PINN)	Use of Hybrid Models (e.g., Physics-informed Deep Learning) and real-time processing.	Limited data quality (gaps, noise) and infrastructure for massive data streams.
Remote Sensing/Vision	Implementation of Drones and AI Vision for inspecting hazardous areas (e.g., furnace refractory)	High initial cost and lack of skilled human capital hinder adoption

LITERATURE REVIEW

1. JFE Technical Report (2020) – Condition Monitoring in Steelmaking with Data Science

- ICT and AI-based condition monitoring (anomaly detection in hot rolling mills): Sensors (visible light cameras, vibration sensors), Wireless LAN (5 GHz band Wi-Fi), Data science/AI models for anomaly detection.
 - Widely studied in Japan (JFE Steel) but limited industrial penetration in India due to cost and infrastructure gaps.

2. Nippon Steel Technical Report (2016) – Equipment Diagnosis Technology

- Condition-based maintenance (CBM), vibration monitoring, ELESMAART portable diagnosis devices, lubricant sampling: Vibration analysis systems, Lubricant contamination analysis tools.
 - Japan has over 40 years of continuous research and deployment while Indian steel plants mainly rely on preventive rather than predictive systems; full CBM is still emerging.

3. AI Approaches for Predictive Maintenance in the Steel Industry (Survey, 2024)

- AI/ML methods for Predictive Maintenance (neural networks, anomaly detection, RUL prediction): Industrial sensors for real-time data, ML/DL frameworks (TensorFlow, PyTorch), High-performance and computing infrastructure.
 - Growing adoption, esp. in EU, China, ArcelorMittal plants.

Continued

4. Data Analysis – Drone-Based Structural Health Monitoring (2024)

- Drone-enabled “fly-by” monitoring with AI, deep learning, and digital twins for structural health monitoring: UAVs with high-resolution cameras/sensors, Computer vision + AI for defect detection, Digital twin platforms.
 - Gaining traction in civil infrastructure (bridges, towers).

5. Applied Sciences (2025) – NLP-Assisted Review of Predictive Maintenanceapplsci

- Hybrid approaches combining signal processing (FFT, wavelets, CEEMDAN), ML/DL (CNNs, LSTMs), and physics-based models: Advanced signal processing software, ML frameworks, IoT sensors for vibration, temperature, current.
 - Hybrid methods discussed in academia; pilot implementations in EU/US.

6. Industry 4.0 Case Study – IoT-based Condition Monitoring (2023)

- IoT sensor nodes for CNC laser cutting machine, dashboard for real-time monitoring, failure alerts using FMEA framework: Wireless IoT sensors, Cloud/centralized dashboards, Data visualization tools.
 - SMEs in EU/UK piloting IoT-enabled condition monitoring.

CHALLENGES

Steel plants are hot, dusty, and corrosive, making sensors and monitoring devices unreliable over time.

Harsh Environments

High Costs

Setting up AI, IoT, or drones and upgrading legacy equipment is very expensive and slows adoption.

There is a shortage of diverse, labeled fault data, making it hard to train accurate AI/ML models.

Data Limitations

Integration Issues

Old, complex machinery in steel plants is difficult to retrofit with modern monitoring technologies.

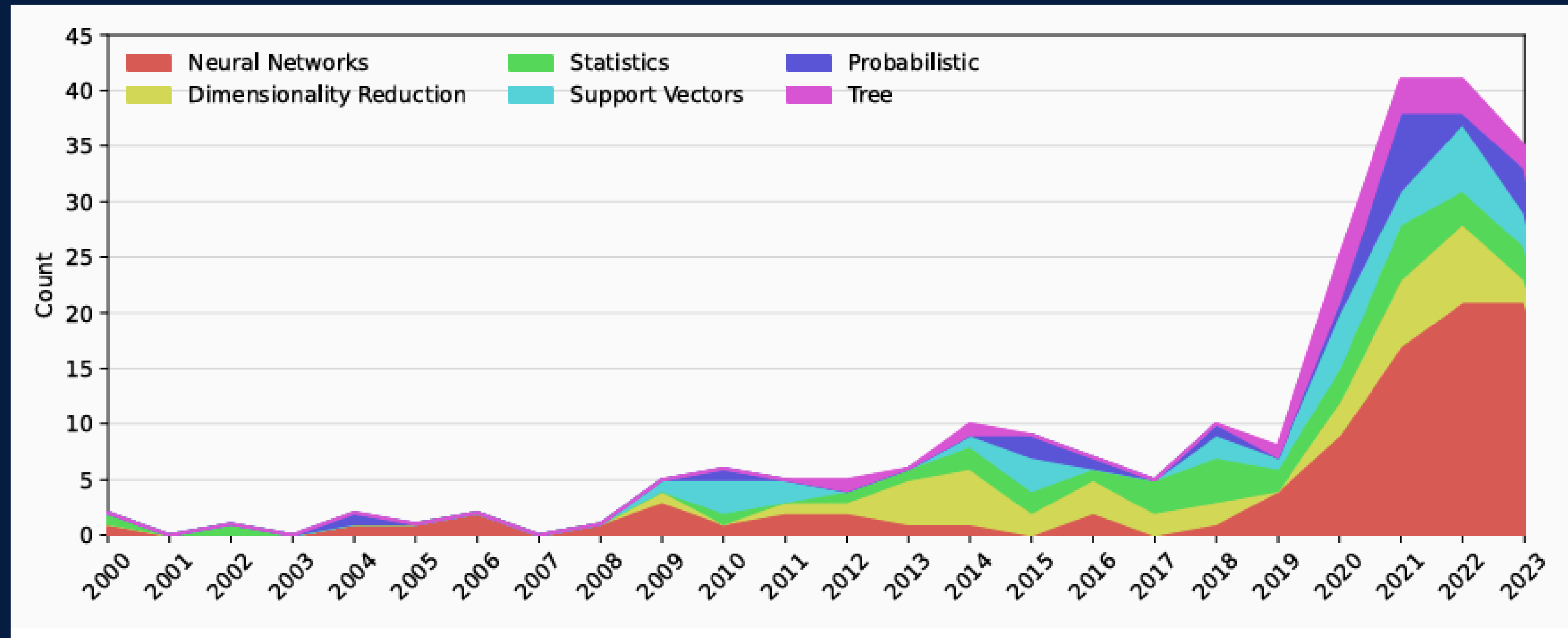
AI models act like black boxes, need heavy computing, and IoT/drones generate noisy, unstable data.

Technical Limits

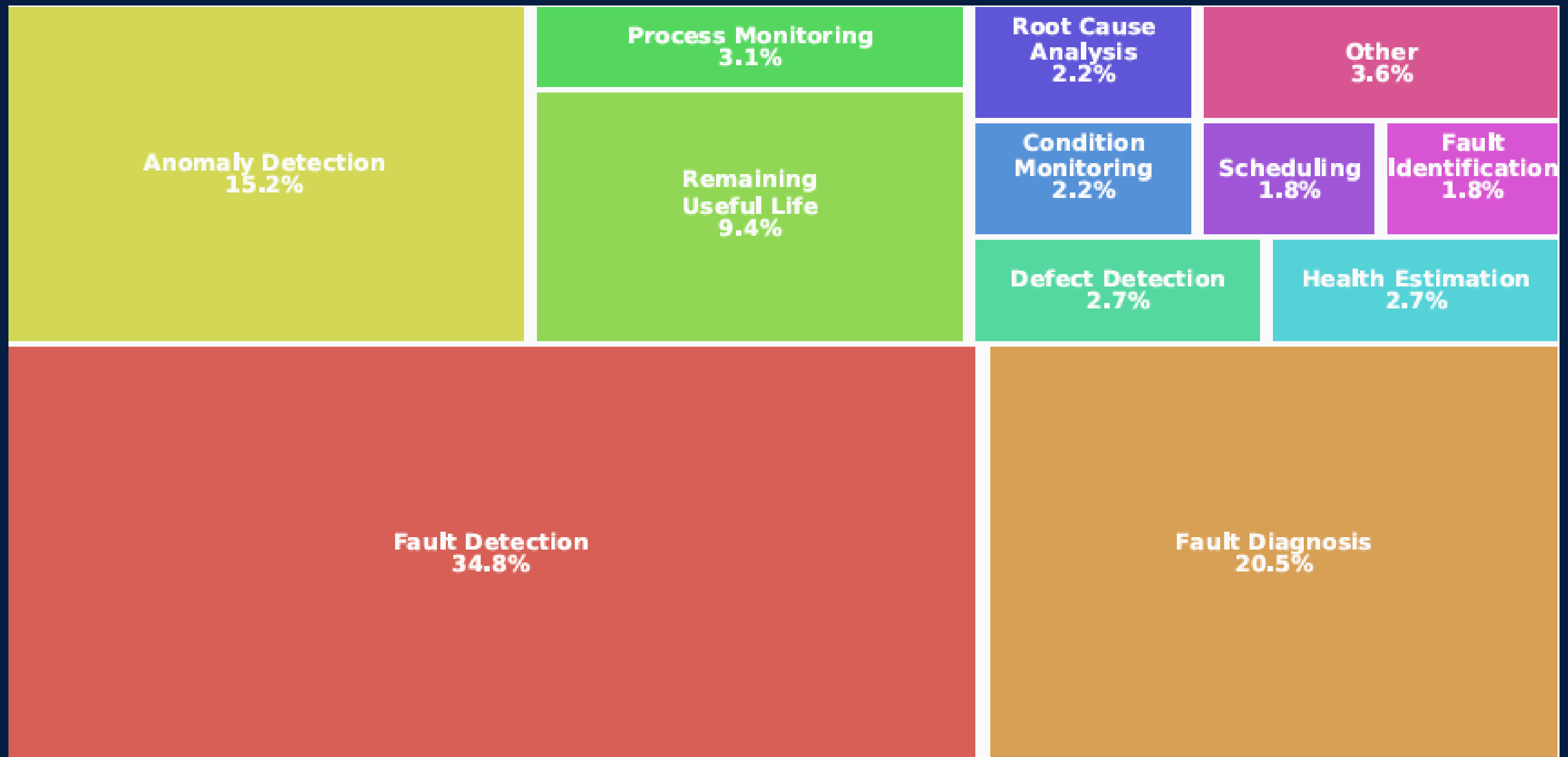
Skills Gap

Advanced systems need trained engineers and data experts, but such skilled staff are often limited.

DIFFERENT ML MODELS IMPLEMENTED IN STEEL INDUSTRIES



This graph shows that predictive maintenance research has grown quickly **after 2015**, with AI methods like **neural networks** becoming the most common.

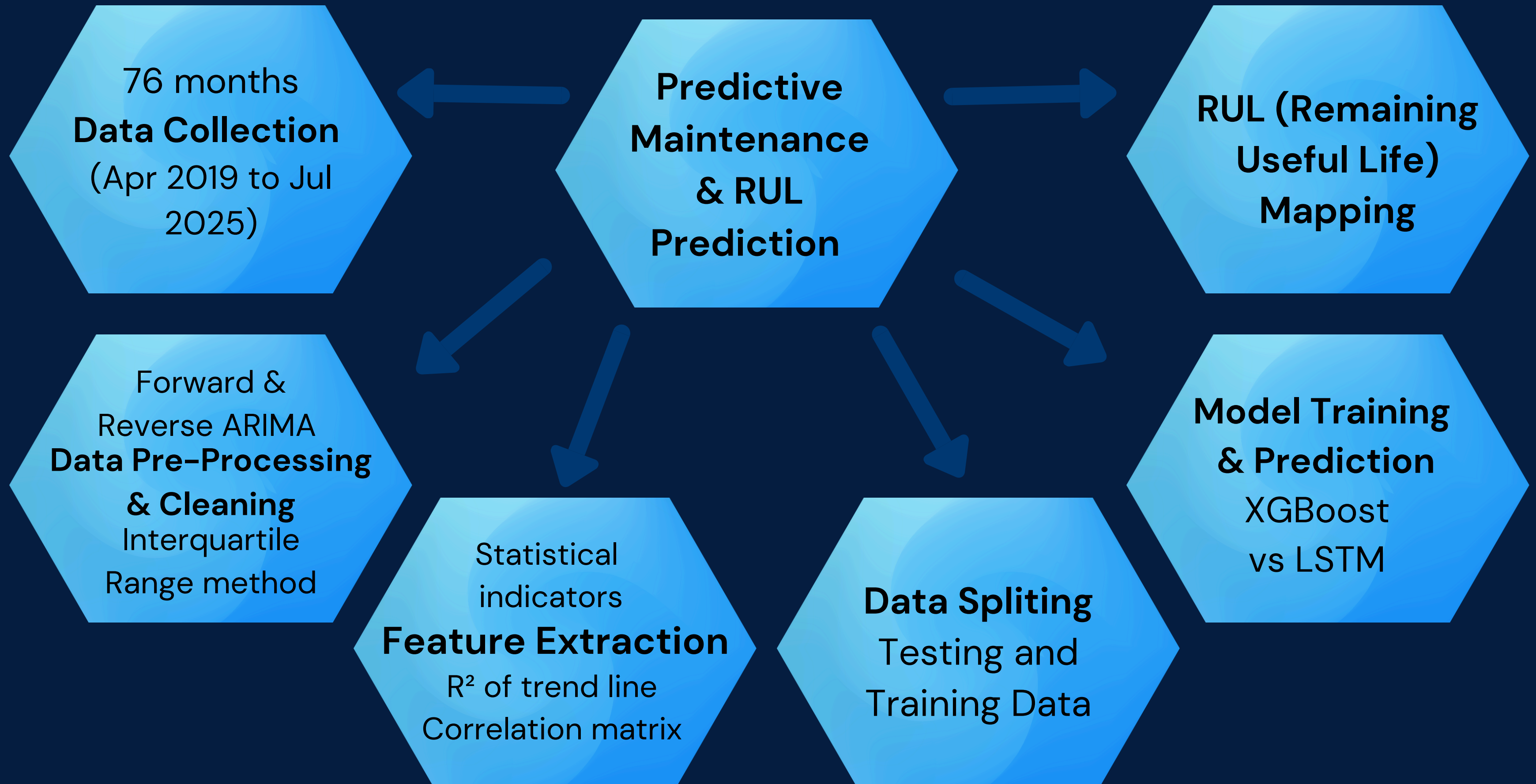


This graph shows where research is focused: mostly on finding faults (detection 35%), understanding them (diagnosis 20%), and spotting unusual behavior (anomaly detection 15%), **but RUL of 9.4% only.**



PROJECT WORK

Workflow Adopted



Step 1 : Understanding the Data

Vibration Data for different Categories

- 76 months data (2019 April to 2025 July)
- 1500 rows – Conveyor (200)
- The raw vibration data is unstable and contains many outliers and sudden spikes. Zero values and null entries make the dataset incomplete.

S.no.	Department	Equipment	Category	Criticality	Total Average	Average last 1Y	Achieved Min	Achieved Max
1	RMHS	1.1:COAL CO	Conveyor	A	2.358948276	2.34	1.181	8.45
2	RMHS	1.2:COAL CO	Conveyor	A	5.077136364	3.3575	2.8	10.52
3	RMHS	1.3:COAL CO	Conveyor	A	5.507439394	5.17875	3.8	9.69
4	RMHS	1.4:COAL CO	Conveyor	A	4.241803279	2.45	2	7.74
5	RMHS	1.5:COAL CO	Conveyor	A	5.230854839	5.616666667	1.9	11.9
6	RMHS	1.6:COAL CO	Conveyor	A	6.019523077	6.512857143	3.8	9.29
7	RMHS	1.7:BOOM CC	Conveyor	A	8.848966667	9.12	3.17	16.3
8	RMHS	1.8:COAL YAF	Conveyor	A	5.804140625	6.45625	2.8	11.8

Avg of best 6	tandard as per IS	Target	Last value	Outliers 2Y	Zero Outliers	Red	Yellow	Green	Status	2019-Apr
1.275166667	4.5	4.5	2.7	1	0	0	0	1	Green	2.18
3.15	4.5	4.5	3.7	47	0	0	0	1	Green	3.83
4.066666667	4.5	4.5	5.7	55	0	0	1	0	Yellow	5.976863
2.324666667	4.5	4.5	3.3	23	0	0	0	1	Green	5.88
2.1	4.5	4.5	6.3	46	0	0	1	0	Yellow	1.92
4.066666667	4.5	4.5	6.4	56	0	0	1	0	Yellow	7.82
4.75	4.5	4.75	5.8	57	0	0	1	0	Yellow	12.6
3.421666667	4.5	4.5	4.5	50	0	0	0	1	Green	7.76
4.8965	4.5	4.8965	5.3	57	0	0	1	0	Yellow	9.99

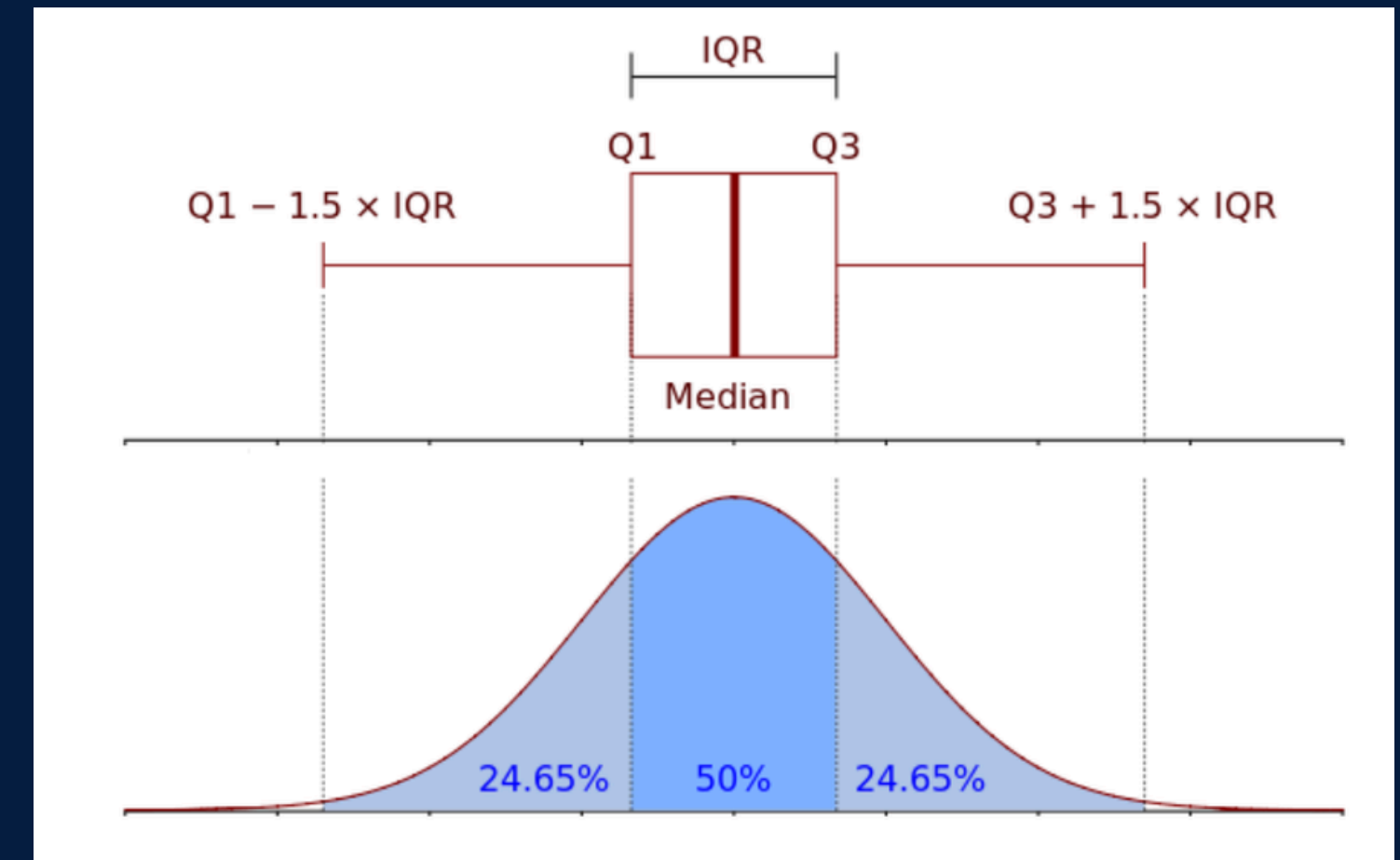
Such noisy data is difficult to train machine learning models on. Hence, preprocessing is essential to clean data, ensuring more reliable patterns and better model performance.

Step 2: PreProcessing

IQR and ARIMA Methods

Interquartile Range (IQR)

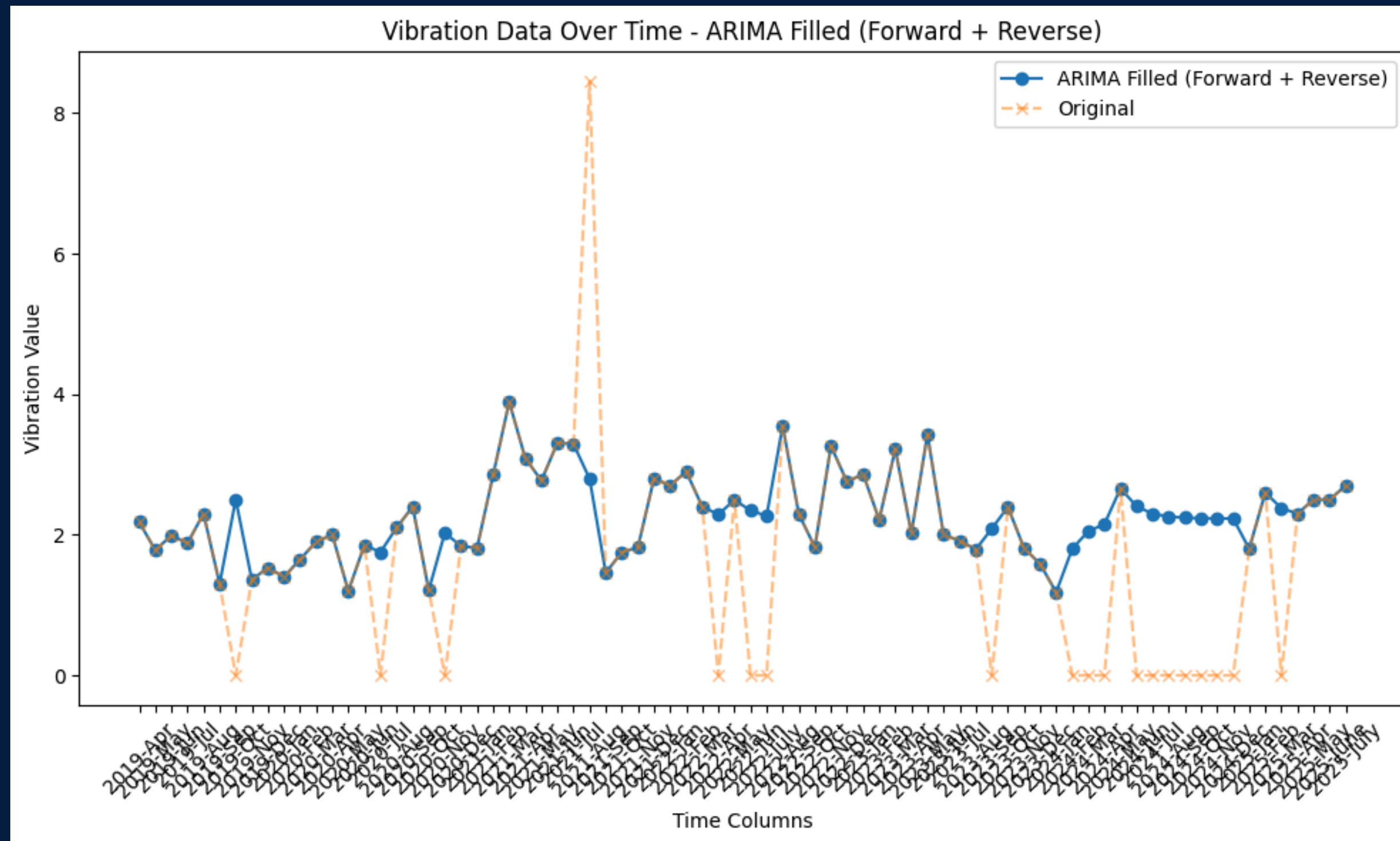
- The IQR is the range between the 25th percentile (Q1) and 75th percentile (Q3) of the data.
- It shows the “middle 50 %” of values.
- Calculate $IQR = Q3 - Q1$.
- Any data point below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is treated as an outlier.



ARIMA (Auto Regressive Integrated Moving Average.)

- Predicts the next value of a time series by combining:
- Auto-regression (AR): looks at how past values affect the next value.
- Integrated (I): uses differences between values to remove long-term trends.
- Moving Average (MA): uses patterns of past prediction errors.

This kept the time series continuous so the RUL models (XGBoost/LSTM) could learn correctly.



Reverse ARIMA

- If missing values are at the start of the series, forward filling cannot handle them.
- So the series is reversed, and ARIMA is applied again to predict missing points using data from the reversed sequence.
- After filling, the sequence is flipped back to the original order.

Step 3: Feature Extraction

13 features are considered.

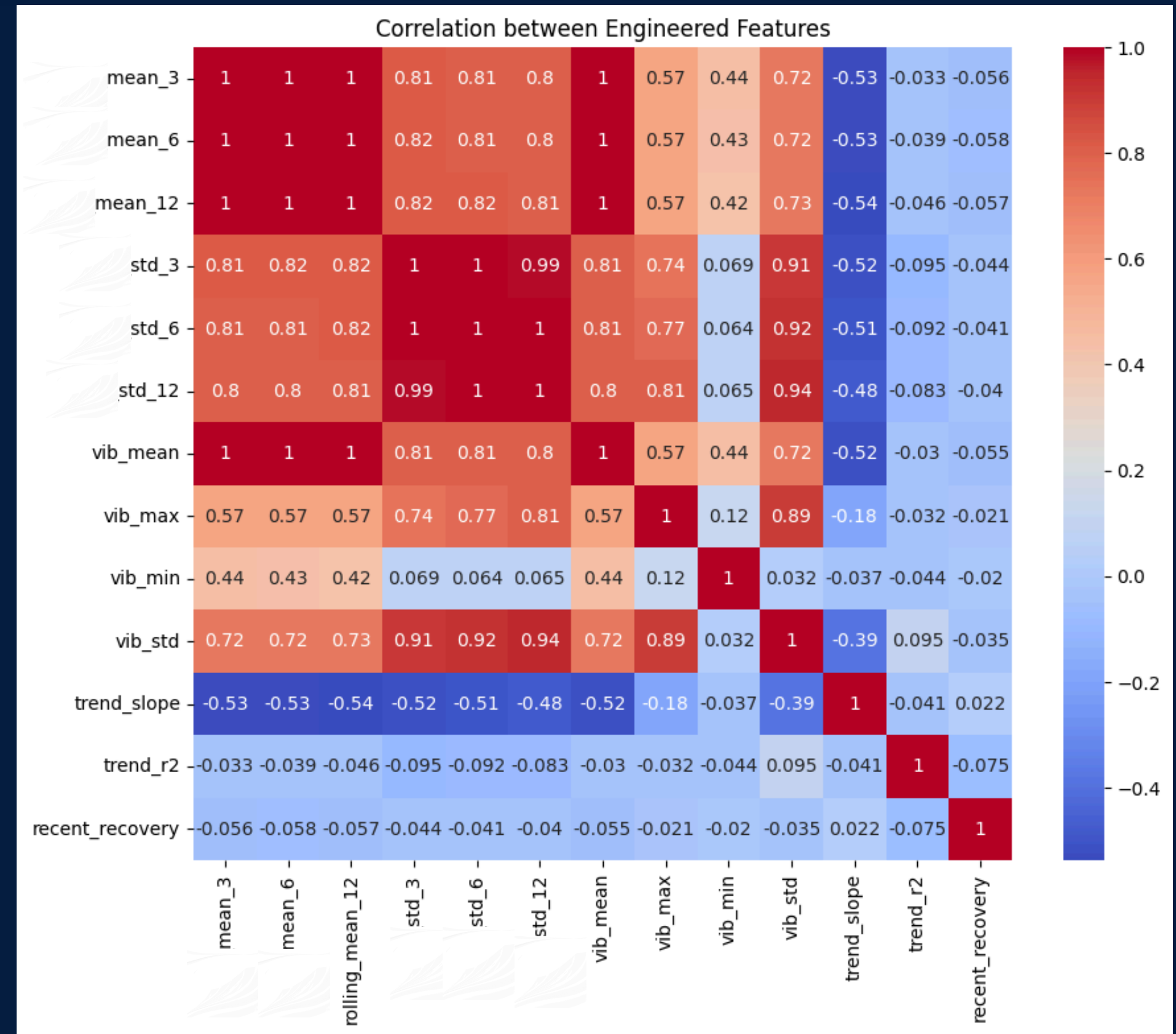
These features summarize raw signals into meaningful indicators:

- Mean features → Baseline load/imbalance.
- Std features → Instability or wear.
- Max/Min → Extreme events.
- Trend features → Long-term machine health.
- Recovery → Detects temporary anomalies.
- After extraction, a correlation matrix is used to check how strongly features are related.
- **High correlation** (close to 1 or -1) means features carry similar information
- **Low correlation** means features capture unique aspects of the vibration data → **valuable for predictive models.**

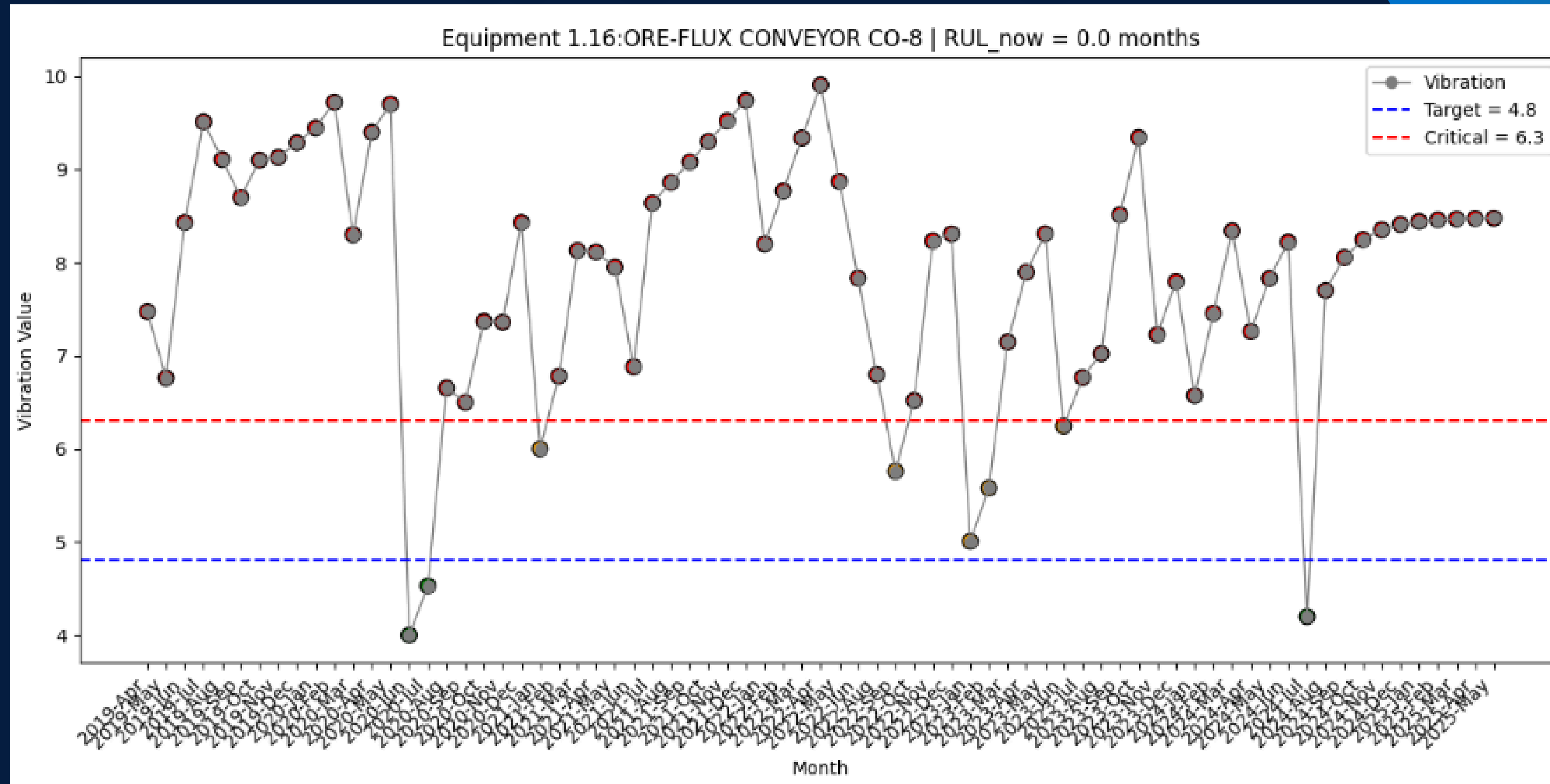
Chosen features:

one mean, one std, vib_max, vib_std, trend_slope, trend_r2, and recent_recovery.

Corelation Matrix



Step 4: RUL Mapping for Existing Data



- A target threshold = 4.8 (blue line) is set, beyond which vibrations indicate potential risk.
- A critical threshold = $1.3 \times \text{target} = 6.3$ (red line) defines severe risk.

- RUL = number of months left until the next Status = 2 (failure condition).
- If the machine is already in failure (status = 2), RUL = 0.
- This calculation is done row by row, month by month, to create a degradation timeline.

For Ore Conveyor 16 Data:

- 2019-Apr → Status = 0 (Healthy)
 - Next failure (status = 2) occurs in 2019-Jun.
 - Hence, RUL(2019-Apr) = 2 months.
- 2019-May → Status = 1 (Warning)
 - Still before 2019-Jun failure.
 - So, RUL(2019-May) = 1 month.
- 2019-Jun → Status = 2 (Failure)
 - Machine already failed.
 - So, RUL(2019-Jun) = 0 months.

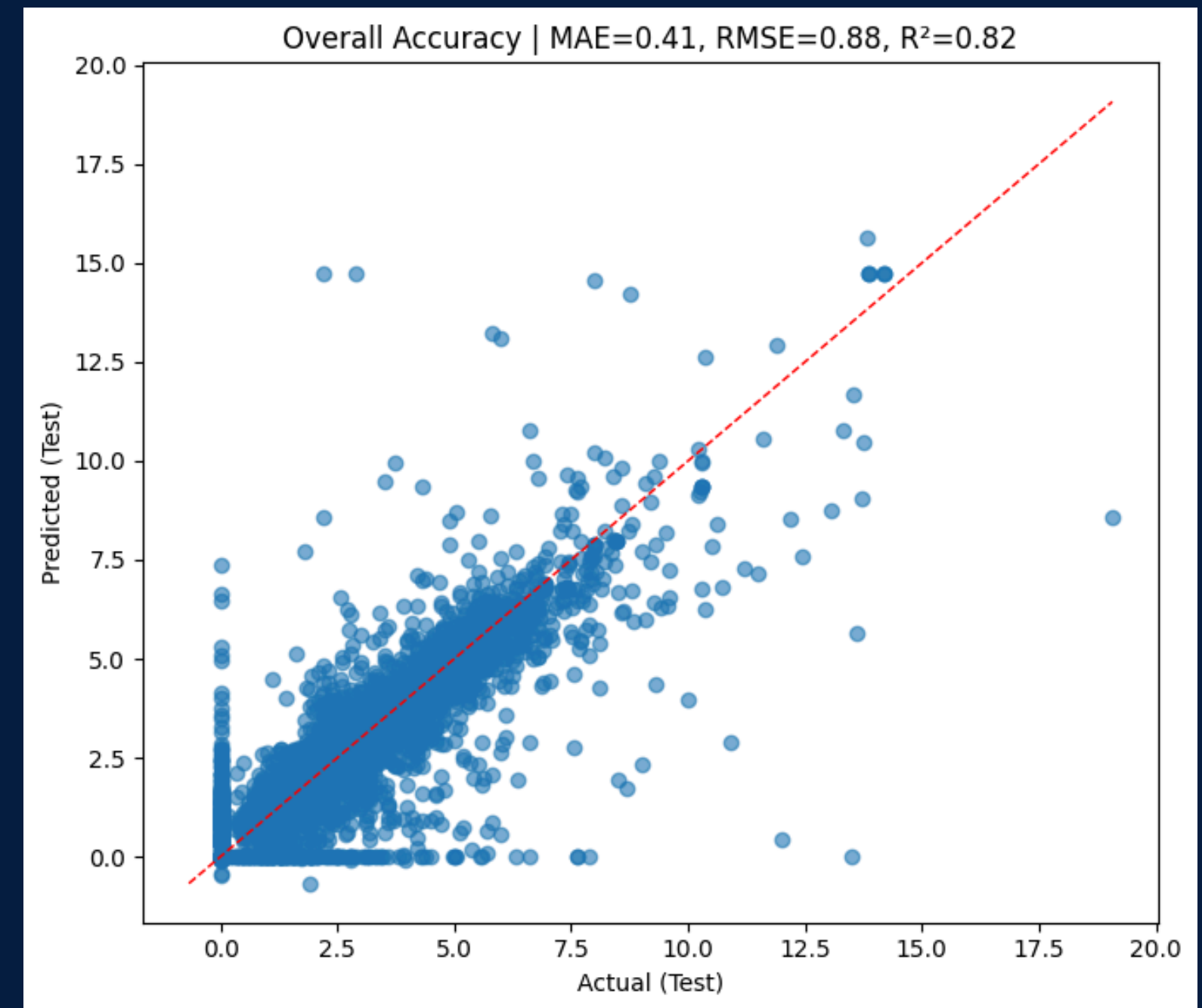
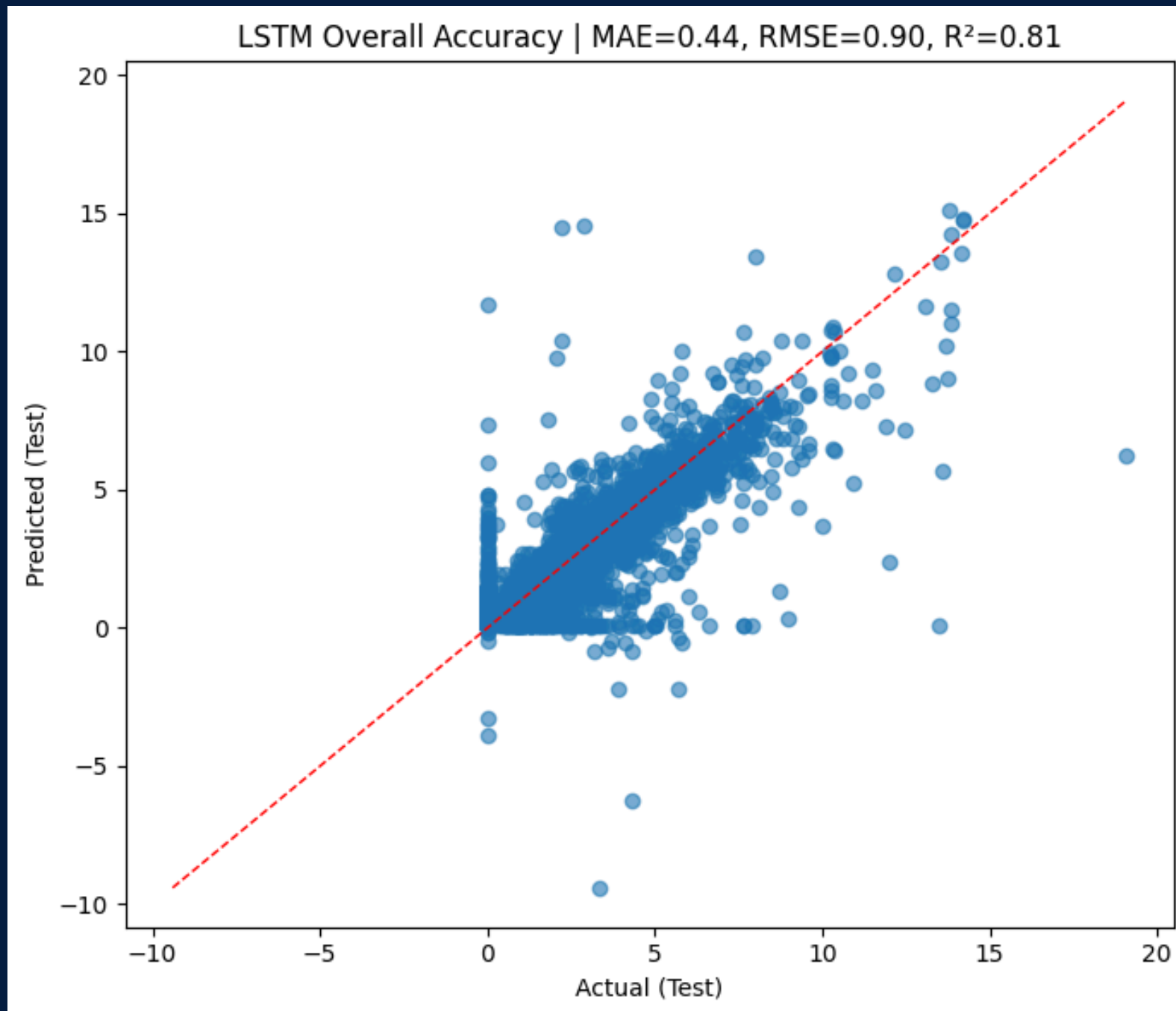
Status_2019-Apr	Status_2019-May	Status_2019-Jun	Status_2019-Jul
0	1	1	2

RUL_2019-Apr	RUL_2019-May	RUL_2019-Jun	RUL_2019-Jul
3	2	1	0

Step 5: Model Training and Testing

80% Training (60 months) and 20% Testing (16 months) – Splitting of data

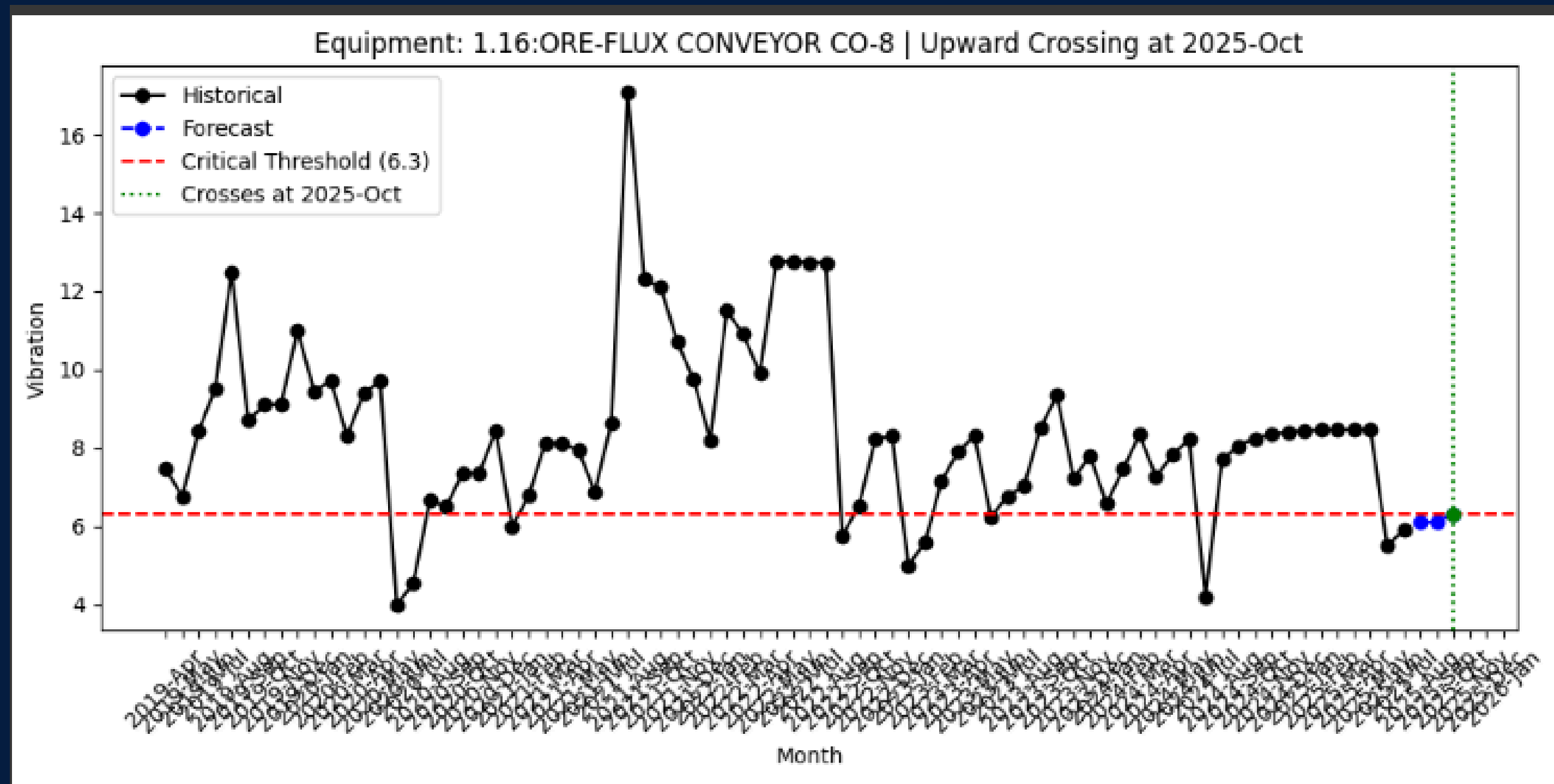
LSTM (Deep Learning) vs XGBoost (Machine Learning)



LSTM (Deep Learning) vs XGBoost (Machine Learning)

Metric	XGBoost	LSTM
MAE (Mean Absolute Error)	0.41	0.55
RMSE (Root Mean Square Error)	0.88	1.15
R ² Score (Coefficient of Determination)	0.82	0.75

Step 6: Execution and Result



This graph shows the predicted vibration trend for the Ore-Flux Conveyor CO-8 over the next few months. 3rd month Failure.

FUTURE ASPECTS

ML Based Monitoring

- Improve the way we fill in missing data by using methods that look at the trend from both past and future readings, not just one side.
- Combine tree models such as XGBoost with deep networks in an **ensemble**, improving accuracy and offering clearer feature-level explanations
- Extracting **time–frequency features** (spectrogram or wavelet-based) that capture early bearing and gearbox faults.
- Use **transfer learning** and synthetic data augmentation to cope with limited labelled failures and improve model robustness.
- Provide **uncertainty estimates** (quantile regression or Bayesian approaches) so maintenance planners can balance risk and cost.

Study : PdM in Mining : > 90% classification accuracy when inference was done on sensor devices and ~99% when using cloud inference.

Drone-Based Monitoring

Use of Unmanned Aerial Vehicles (UAVs) or drones (swarm) with sensors, high resolution cameras, thermal imaging, LiDAR, infrared.

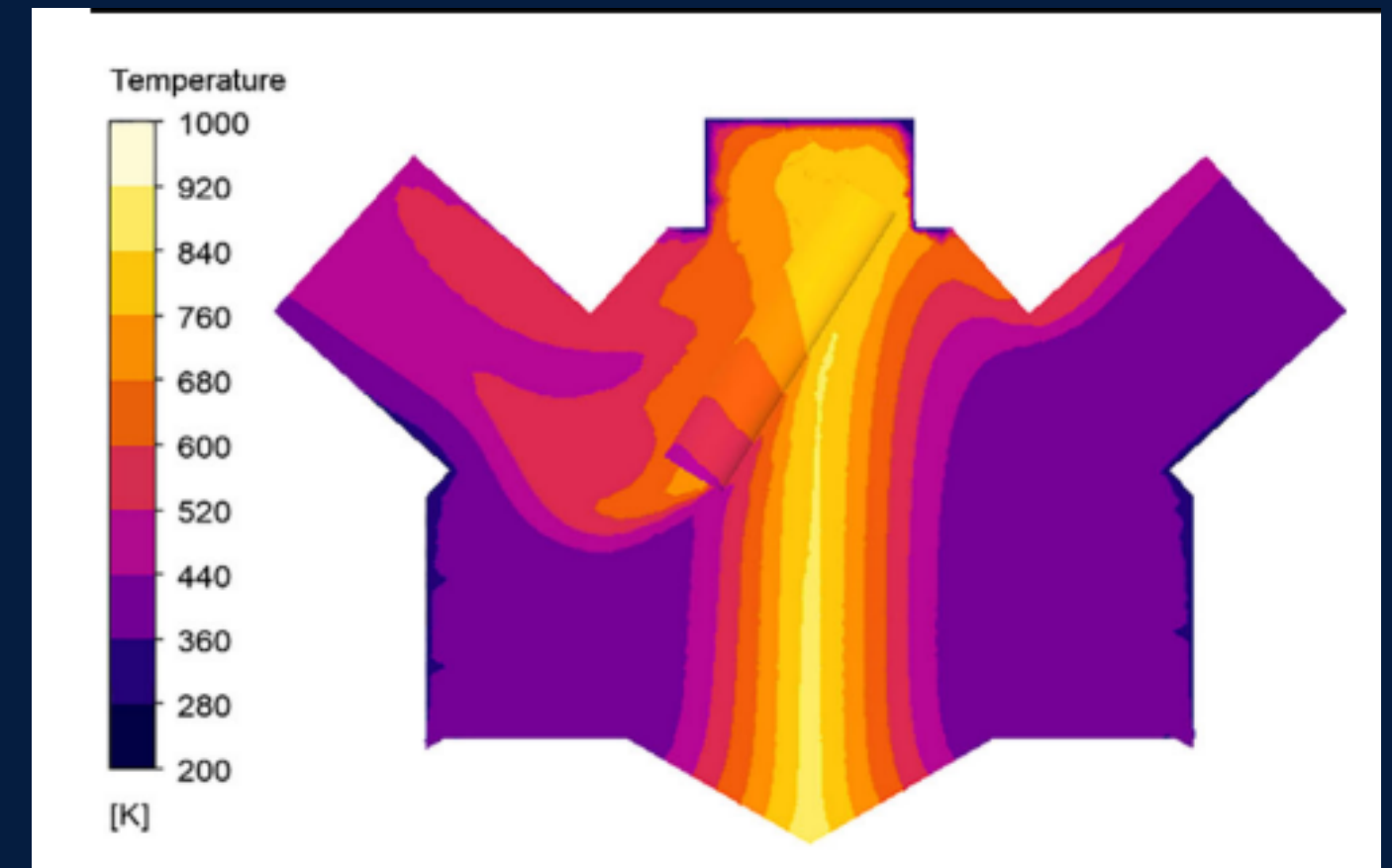
UAV Formation Control: Drones maintain relative positions using formation frame (x_F, y_F, z_F) and GPS/RTK for accurate localization

Path Planning & Navigation: Optimized flight paths generated via Particle Swarm Optimization (PSO), A or RRT algorithms ensure full surface coverage.

Chinese Study on BF Chute Thermograms

Thermograms : PSNR ≈ 33.801 , SSIM ≈ 0.980 , MAE ≈ 0.013

Europe: 80% less inspection time compared to traditional methods.



Thermal image generation for blast furnace chute based on Generative Adversarial Network (GAN)

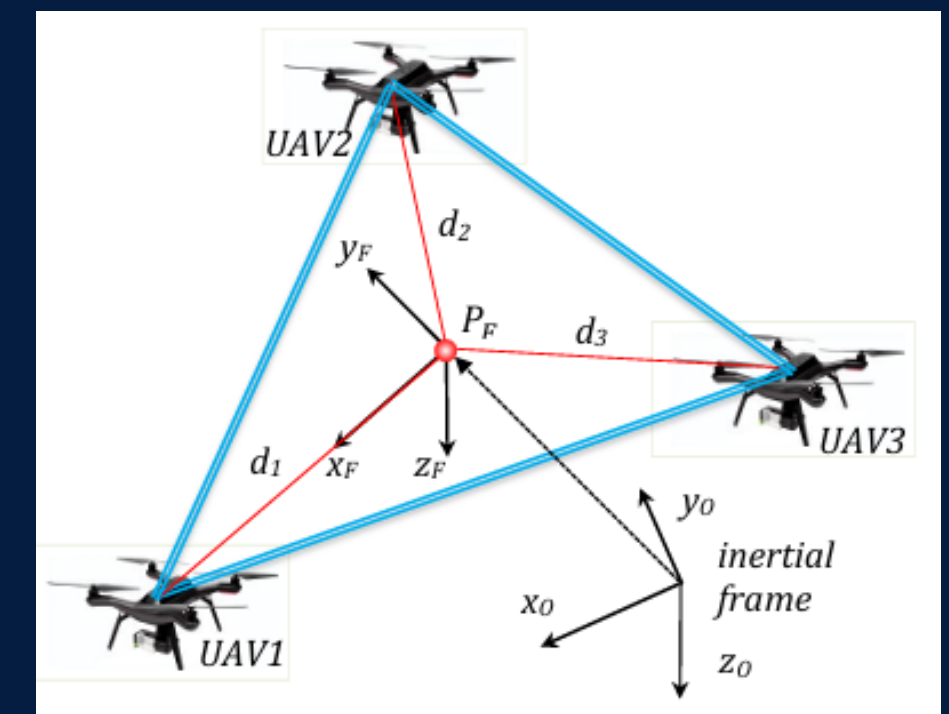


Figure : Inertial and formation frames in UAV

SUMMARY

- Steelmaking involves complex equipment (blast furnace, BOF, ladle, rolling mills) prone to failures.
- Equipment breakdowns cause huge costs – global downtime losses reach ~\$1.5 trillion/year.
- Predictive Maintenance (PdM) reduces unplanned downtime by up to 50% and maintenance costs by 18–25%.
- Literature shows wide global adoption of AI, IoT, drones, and digital twins for monitoring, but India lags due to cost and infrastructure gaps.
- Main challenges: harsh environments, high costs, data limitations, technical issues, and shortage of skilled staff.
- Project work focused on vibration prediction and RUL Calculations, showing how PdM can prevent dangerous breakdowns and accidents.
- Future aspects: drones, AI-based thermal imaging, and optimized navigation to speed up inspections and improve safety.

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THANK YOU!